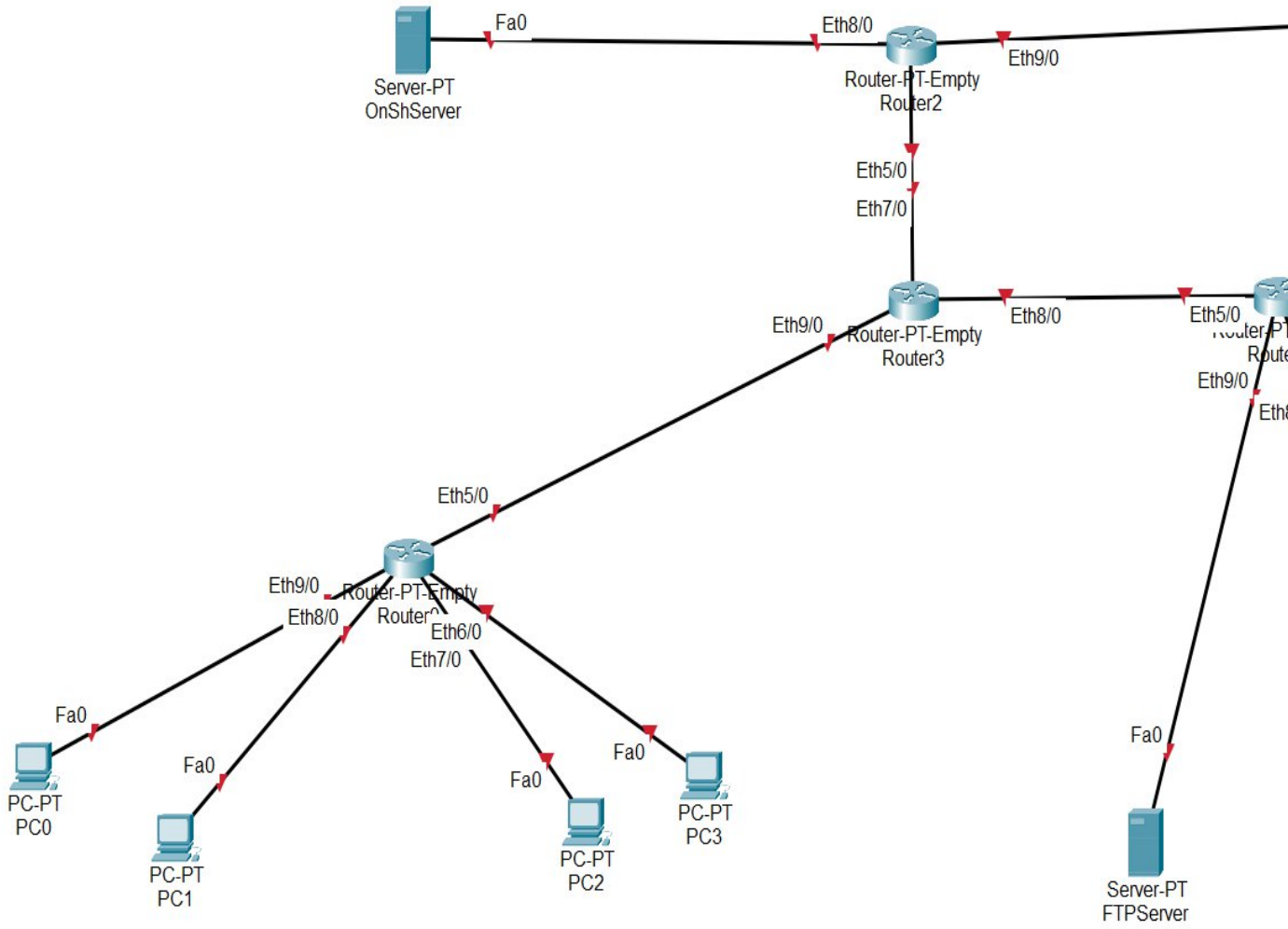
Задание 1. Формирование адресного пространства для головного офиса

Назначение подсети	**Подсеть (IP/маска)**	**Диапазон IP-адресов (IP-IP)**	**Широковещательный адрес (IP)**
Маркетинг	10.0.0.0/11	10.0.0.1 – 10.31.255.254	10.31.255.255
Финансы	10.32.0.0/11	10.32.0.1 – 10.63.255.254	10.63.255.255
Сотрудники IT	10.64.0.0/11	10.64.0.1 – 10.95.255.254	10.95.255.255
Сотрудники ИБ	10.96.0.0/11	10.96.0.1 – 10.127.255.254	10.127.255.255
Электронная почта	10.128.0.0/12	10.128.0.1 – 10.143.255.254	10.143.255.255
Онлайн-магазин	10.144.0.0/12	10.144.0.1 – 10.159.255.254	10.159.255.255
FTP-сервер	10.160.0.0/12	10.160.0.1 – 10.175.255.254	10.175.255.255
1С:Предприятие	10.176.0.0/12	10.176.0.1 – 10.191.255.254	10.191.255.255
DNS-сервер	10.192.0.0/12	10.192.0.1 – 10.207.255.254	10.207.255.255
Service Desk	10.208.0.0/12	10.208.0.1 – 10.223.255.254	10.223.255.255

Задание 2. Формирование политики доступа

IP/Сеть-источник	**IP/Сеть назначения**	**Транспортный протокол**	**Порт**	**Прикладной протокол**
10.0.0.0/11, 10.32.0.0/11, 10.64.0.0/11, 10.96.0.0/11.	10.192.0.0/12	UDP	53	DNS
10.0.0.0/11, 10.64.0.0/11, 10.96.0.0/11.	10.144.0.0/12	TCP	443	HTTPS
10.32.0.0/11, 10.64.0.0/11, 10.96.0.0/11.	10.176.0.0/12	TCP	443	HTTPS
10.0.0.0/11, 10.32.0.0/11, 10.64.0.0/11, 10.96.0.0/11.	10.128.0.0/12	TCP	25	SMTP
10.0.0.0/11, 10.32.0.0/11, 10.64.0.0/11, 10.96.0.0/11.	10.208.0.0/12	TCP	443	HTTPS
10.32.0.0/11, 10.64.0.0/11, 10.96.0.0/11.	10.160.0.0/12	TCP	21	FTP

Задание 3. Сборка стенда



Задание 4. Формирование адресного пространства

Device	Port	IP/mask
PC0	Fa0	10.0.0.1/11
PC1	Fa0	10.32.0.1/11
PC2	Fa0	10.64.0.1/11
PC3	Fa0	10.96.0.1/11
FTP-Server	Fa0	10.160.0.1/12
DNS-Server	Fa0	10.192.0.1/12
Online Shop-Server	Fa0	10.144.0.1/12
Mail-Server	Fa0	10.128.0.1/12
Router0	Eth9/0	10.31.255.254/11
Router0	Eth8/0	10.63.255.254/11
Router0	Eth7/0	10.95.255.254/11
Router0	Eth6/0	10.127.255.254/11
Router0	Eth5/0	192.168.1.1/30
Router1	Eth9/0	10.175.255.254/12
Router1	Eth8/0	10.207.255.254/12
Router1	Eth5/0	192.168.1.9/30
Router2	Eth9/0	10.143.255.254/12

Router2	Eth8/0	10.159.255.254/12
Router2	Eth5/0	192.168.1.5/30
Router3	Eth9/0	192.168.1.2/30
Router3	Eth8/0	192.168.1.10/30
Router3	Eth7/0	192.168.1.6/30

Задание 5. Настройка статической маршрутизации

- Router0

```
Router0#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.1.2 to network 0.0.0.0

     10.0.0.0/11 is subnetted, 4 subnets
C       10.0.0.0 is directly connected, Ethernet9/0
C       10.32.0.0 is directly connected, Ethernet8/0
C       10.64.0.0 is directly connected, Ethernet7/0
C       10.96.0.0 is directly connected, Ethernet6/0
     192.168.1.0/30 is subnetted, 1 subnets
C       192.168.1.0 is directly connected, Ethernet5/0
S*   0.0.0.0/0 [1/0] via 192.168.1.2
```

- Router1

```
Router1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.1.10 to network 0.0.0.0

     10.0.0.0/12 is subnetted, 2 subnets
C       10.160.0.0 is directly connected, Ethernet9/0
C       10.192.0.0 is directly connected, Ethernet8/0
     192.168.1.0/30 is subnetted, 1 subnets
C       192.168.1.8 is directly connected, Ethernet5/0
S*   0.0.0.0/0 [1/0] via 192.168.1.10
```

- Router2

```
Router2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.1.6 to network 0.0.0.0

     10.0.0.0/12 is subnetted, 2 subnets
C       10.128.0.0 is directly connected, Ethernet9/0
C       10.144.0.0 is directly connected, Ethernet8/0
     192.168.1.0/30 is subnetted, 1 subnets
C       192.168.1.4 is directly connected, Ethernet5/0
S*   0.0.0.0/0 [1/0] via 192.168.1.6
```

- Router3

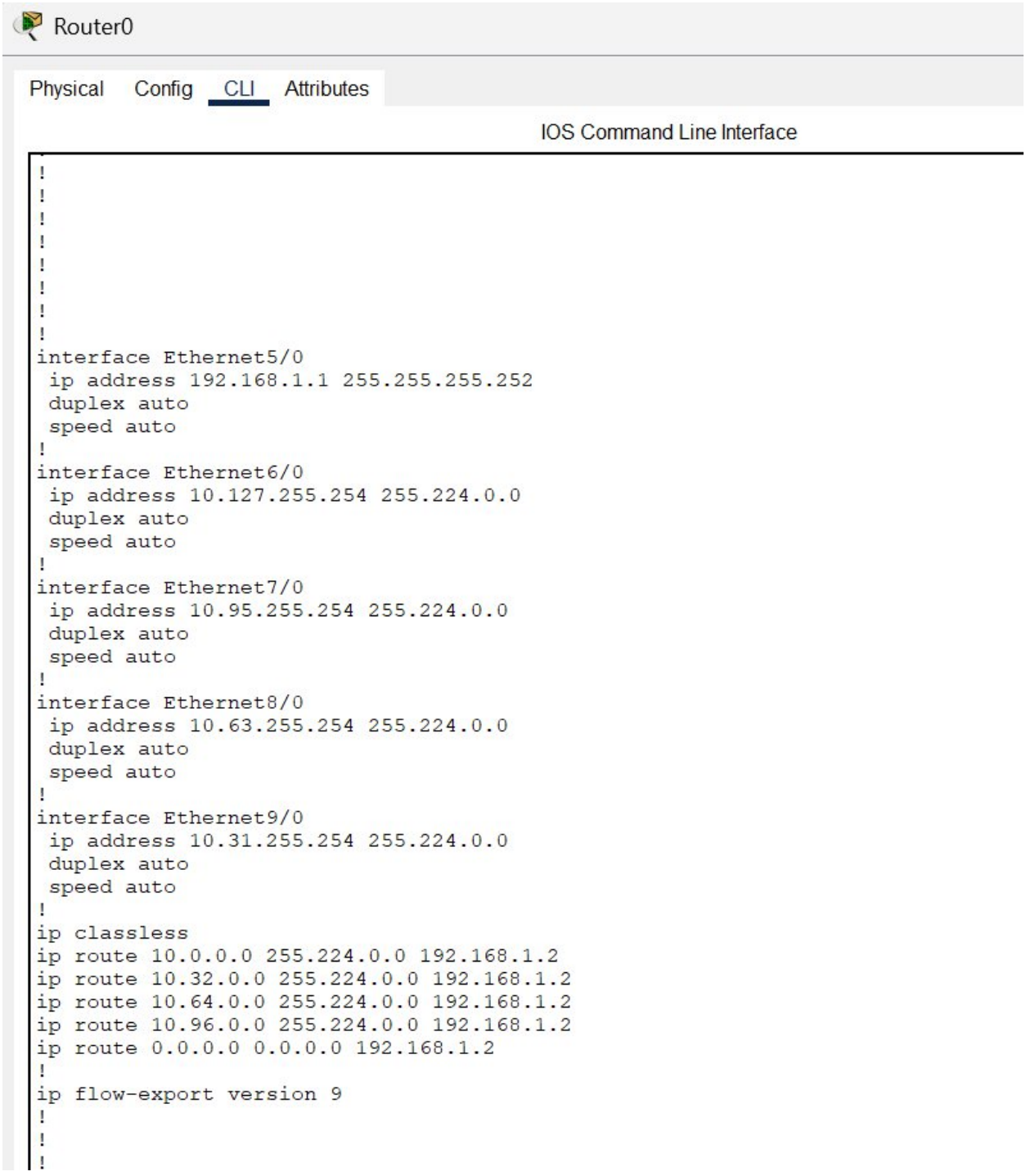
```
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

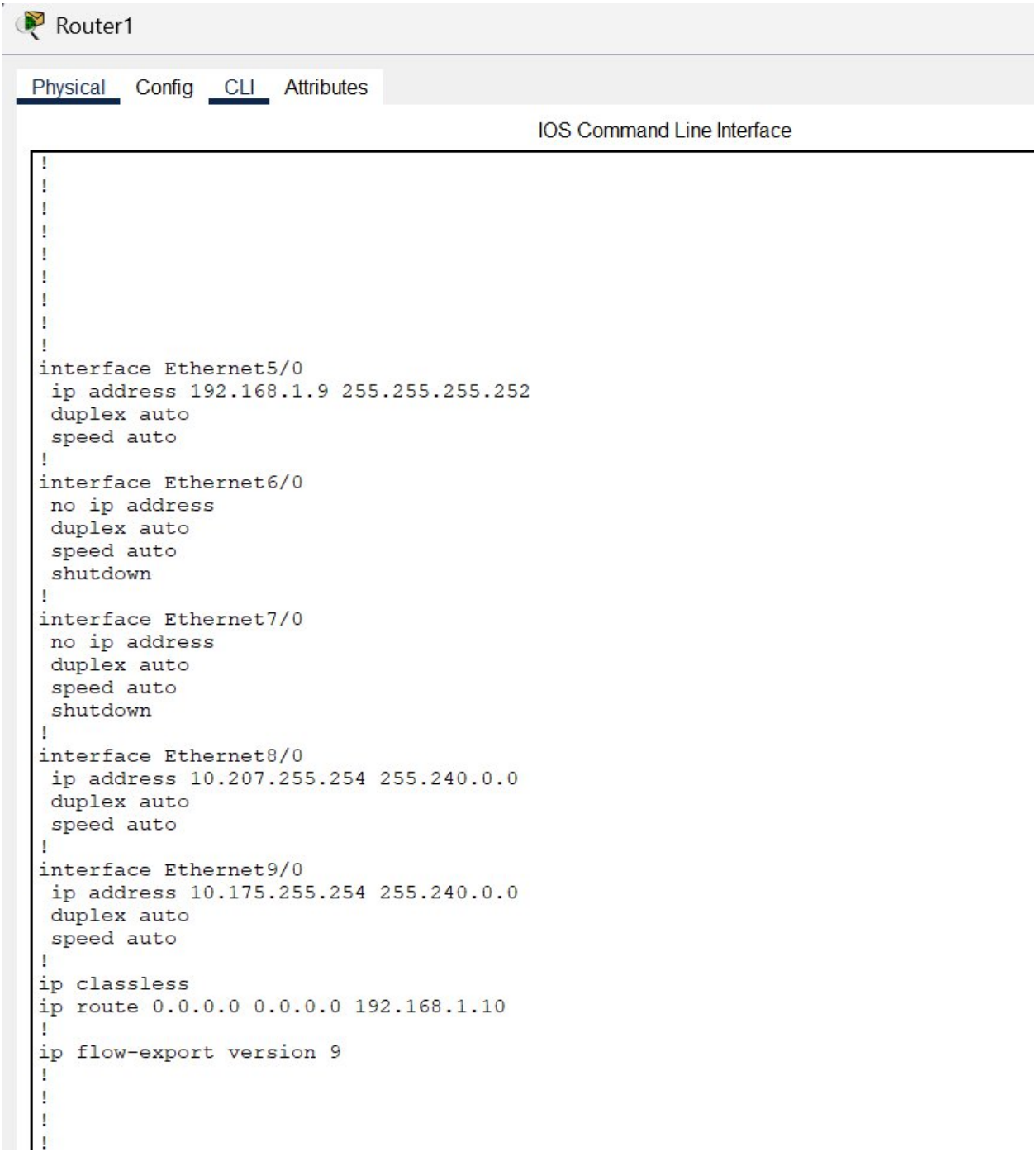
     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
S       10.0.0.0/11 [1/0] via 192.168.1.1
S       10.32.0.0/11 [1/0] via 192.168.1.1
S       10.64.0.0/11 [1/0] via 192.168.1.1
S       10.96.0.0/11 [1/0] via 192.168.1.1
S       10.128.0.0/12 [1/0] via 192.168.1.5
S       10.144.0.0/12 [1/0] via 192.168.1.5
S       10.160.0.0/12 [1/0] via 192.168.1.9
S       10.192.0.0/12 [1/0] via 192.168.1.9
     192.168.1.0/30 is subnetted, 3 subnets
C       192.168.1.0 is directly connected, Ethernet9/0
C       192.168.1.4 is directly connected, Ethernet7/0
C       192.168.1.8 is directly connected, Ethernet8/0
```

Задание 6. Рабочая конфигурация

- Router0



- Router1



- Router2

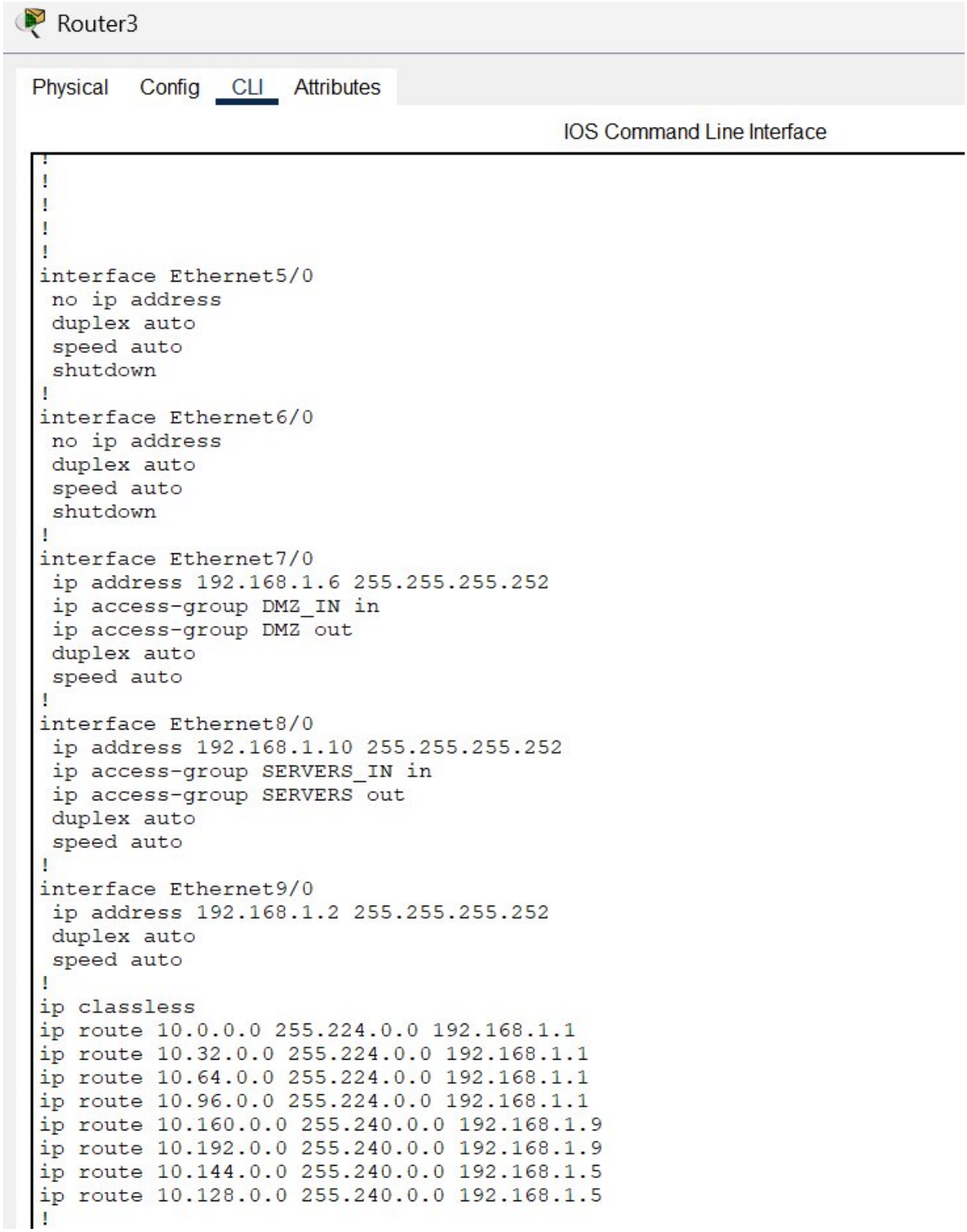
Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
!
!
!
!
!
!
!
!
interface Ethernet5/0
 ip address 192.168.1.5 255.255.255.252
 duplex auto
 speed auto
!
interface Ethernet6/0
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Ethernet7/0
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Ethernet8/0
 ip address 10.159.255.254 255.240.0.0
 duplex auto
 speed auto
!
interface Ethernet9/0
 ip address 10.143.255.254 255.240.0.0
 duplex auto
 speed auto
!
ip classless
ip route 0.0.0.0 0.0.0.0 192.168.1.6
!
ip flow-export version 9
!
!
!
!
!
!
```

- Router3


Router3
Physical
Config
CLI
Attributes
IOS Command Line Interface
!
!
!
!
!
interface Ethernet5/0
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Ethernet6/0
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Ethernet7/0
 ip address 192.168.1.6 255.255.255.252
 ip access-group DMZ_IN in
 ip access-group DMZ out
 duplex auto
 speed auto
!
interface Ethernet8/0
 ip address 192.168.1.10 255.255.255.252
 ip access-group SERVERS_IN in
 ip access-group SERVERS out
 duplex auto
 speed auto
!
interface Ethernet9/0
 ip address 192.168.1.2 255.255.255.252
 duplex auto
 speed auto
!
ip classless
ip route 10.0.0.0 255.224.0.0 192.168.1.1
ip route 10.32.0.0 255.224.0.0 192.168.1.1
ip route 10.64.0.0 255.224.0.0 192.168.1.1
ip route 10.96.0.0 255.224.0.0 192.168.1.1
ip route 10.160.0.0 255.240.0.0 192.168.1.9
ip route 10.192.0.0 255.240.0.0 192.168.1.9
ip route 10.144.0.0 255.240.0.0 192.168.1.5
ip route 10.128.0.0 255.240.0.0 192.168.1.5
!

```
!
ip access-list extended DMZ
 permit tcp any 10.128.0.0 0.15.255.255 eq smtp
 permit tcp 10.0.0.0 0.31.255.255 10.144.0.0 0.15.255.255 eq 443
 permit tcp 10.64.0.0 0.31.255.255 10.144.0.0 0.15.255.255 eq 443
 permit tcp 10.96.0.0 0.31.255.255 10.144.0.0 0.15.255.255 eq 443
 permit icmp any any
ip access-list extended DMZ_IN
 permit icmp any any
 permit tcp 10.128.0.0 0.15.255.255 any established
 permit tcp 10.144.0.0 0.15.255.255 any established
ip access-list extended SERVERS
 permit udp any 10.192.0.0 0.15.255.255 eq domain
 permit tcp 10.32.0.0 0.31.255.255 10.160.0.0 0.15.255.255 eq ftp
 permit tcp 10.64.0.0 0.31.255.255 10.160.0.0 0.15.255.255 eq ftp
 permit tcp 10.96.0.0 0.31.255.255 10.160.0.0 0.15.255.255 eq ftp
 permit icmp any any
ip access-list extended SERVERS_IN
 permit icmp any any
 permit tcp 10.160.0.0 0.15.255.255 any established
 permit udp 10.192.0.0 0.15.255.255 any
!
!
!
!
```

Комментарии

Не понял зачем show run на других роутерах, если все настройки на router3